Lecture 5: The Doppler effect and Rapidity

Based on Morin's *Introduction to Classical Mechanics*, Ch. 11.8–11.9

Outline

1. Recap lecture 4
2. The Doppler effect
3. Rapidity

1 Recap Lecture 4

We showed that we can define a frame-independent quantity, the spacetime interval (or interval, according to Morin):

$$\boxed{\Delta s^2 = c^2 \Delta t^2 - \Delta d^2. \qquad (1)}$$

We showed that $c\Delta t \geq \Delta s$. The formal definition of the timelike spacetime interval states that it is c times the time between two events as measured by an inertial clock present at both events. Because the inertial clock follows a unique worldline, the spacetime interval is unique for a given pair of events (hence it is frame-independent). A geometrical analogy is the distance in Euclidean space.

Besides the spacetime interval, we also derived an expression for proper time:

$$\boxed{\Delta \tau = \int_{t_A}^{t_B} \sqrt{1 - \frac{|\vec{v}|^2}{c^2}}\, dt. \qquad (2)}$$

The formal definition for proper time is the time between two events as measured by a single clock present at both events. Since this clock does not have to be inertial, $c\Delta\tau$ is not equal to the spacetime interval, unless that single clock follows an inertial path (i.e., a straight line in a spacetime diagram). The geometrical equivalent is the path length: the time measured on the clock depends on the precise worldline the clock follows between the two events. While this is a geometrical analogy, the hierarchy is the opposite $c\Delta t \geq \Delta s \geq c\Delta\tau$.

Next, we introduced spacetime diagrams. They allow us to improve our intuition for simple 2D problems. For SR diagrams, it does help to work in SR units (although it is possible to simply label the time axis as ct as Morin does). In that case, we can treat time and space on equal footing and use units that are identical on both axes. Two observer diagrams are useful when considering scenarios where we have to compare two frames. We derived how to draw the axes of the frame S' moving with speed u in the positive x

direction, with $\beta \equiv u/c$, and how to put ticks on these axes. We showed how we can use the two observer diagram to understand length contraction.

1.1 The pole and barn paradox

To show how two observer diagrams can help understand perhaps contradicting statements, we will consider the pole and barn paradox. In this hypothetical scenario, we consider two observers. One observer is running with a pole of proper length $L'_p = 10$ light nanoseconds (lns, SR units) past a barn of proper length $L_{\text{barn}} = 8$ lns[1]. The runner is running with a velocity $\beta = 3/5c$ (pretty fast, yes, it's a paradox). Another observer, who is at rest w.r.t. the runner, sees the runner passing by and the pole contracted $L_p = \gamma^{-1} L'_p = 8$ lns. The pole fits *in* the barn. The runner, on the other hand, passes by the barn at high speed and sees the length of the barn contracted: $L'_{\text{barn}} = \gamma^{-1} L_{\text{barn}} = 6.4$ lns. The runner concludes, the pole does not fit in the barn! Who is right?

The solution to this paradox requires defining length in terms of events! In that case, there is no such general thing as the pole 'fitting' the barn because that relies on events being simultaneous (which depends on coordinate time and thus on the reference frame).

The solution can be understood if we draw a two observer diagram. We show the steps how to draw such a diagram in fig. 1. The main point is that both are right from the perspective of their frames, since what is simultaneous in one frame is not simultaneous in another. Since an object's length in a given inertial frame is defined to be the distance between any two simultaneous events that occur at its ends, they will not agree, but that is fine and fully consistent with SR.

There is a large number of problems that are better understood if you draw a (two observer) spacetime diagram. Morin has many questions, and I encourage you to take a look. For example, you can also derive the rear clock ahead effect using a spacetime diagram. You can find the solution to this problem in one of the exams.

An even more challenging paradox is the skateboard and pothole example. In that example, there is a skateboard (1lns) moving to the right, and a pothole moving up (1lns). Does the skateboard fall through the pothole? If you send me a solution, you will receive a reward.

2 The Doppler effect

2.1 Longitudinal effect

In previous lectures, we have discussed the difference between seeing and observing. The latter refers to when something has actually happened, as can be read off from a lattice with a set of synchronized clocks (the value of t on the coordinate axis). Seeing refers to when the light that comes off an event actually hits your eye. The Doppler effect cares about seeing.

Consider a source that flashes at frequency ν' (in its own frame S') while moving away from you with (constant) speed v. Here v is the source's physical speed relative to the observer in this one-object setup; it is not the boost notation used for a general transformation. The relativistic Doppler effect has two contributions:

[1]Recall, proper length is the length of an object measured at rest.

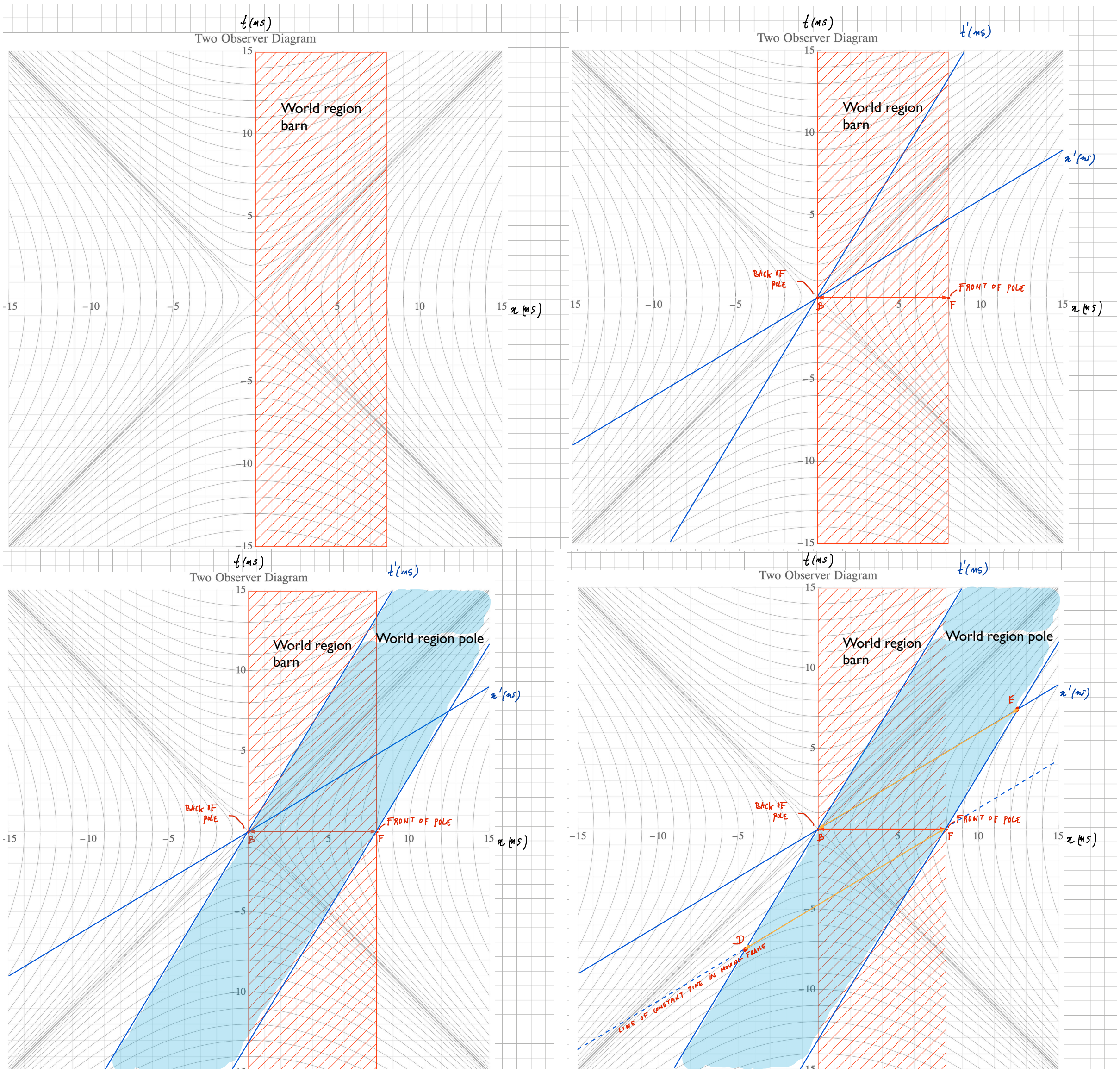

Figure 1: **Top left:** two observer diagram and the worldregion of the barn (8ns) in the frame at rest. **Top Right:** adding x' and t' coordinate axes for the S' frame moving with $3/5c$. **Bottom left:** the pole fits exactly in the barn according to the observer at rest, which is clear from the intersection of the two worldregions (barn and pole) as measured along the spatial coordinate axis x. **Bottom right:** the runner disagrees. In the runner's frame, simultaneous events are parallel to the x' axis, and from S' the two events that are simultaneous show that the pole does not fit in the barn (the pole's front exits the barn before the rear enters the barn, which defines the length of the barn compared to the pole).

- Time dilation: There is more time between flashes in your frame (S), which means that the frequency ν in your frame should go down.
- Classic Doppler: similar to sound, if a source is moving towards you, there is less distance to travel to you (and more if it is moving away). This effect increases the frequency when a source is moving towards you, and reduces it when it is moving away. In the case of sound, you are familiar with a car passing by at high speed, or a siren on an ambulance. The sound will move from a high pitch to a low pitch as the source passes by.

To obtain an actual expression for the Doppler effect, we first note that the time interval between the emissions as measured in frame S' can be expressed as $\Delta t' = 1/\nu'$, i.e., one over the frequency of the flashes as measured in frame S'. We can apply time dilation to find that the time interval between emissions in frame S is given by $\Delta t = \gamma \Delta t'$. Now this is not the time between the flashes as seen by the observer, since the source is moving, and depending on whether the source is moving away or towards you, the distance between you and the source has changed in between emissions. Specifically, the source has travelled $v\Delta t = v\gamma\Delta t'$ in between emissions. So the time ΔT between flashes as seen by the observer in S is given by the difference in travel distance between the two emissions, divided by the speed of a flash (c), i.e.

$$\Delta T = \frac{1}{c}(c+v)\gamma\Delta t' = \frac{1+\beta}{\sqrt{1-\beta^2}}\Delta t' = \sqrt{\frac{1+\beta}{1-\beta}}\nu'^{-1}, \tag{3}$$

where $\beta = v/c$. The frequency of the observed sources is therefore

$$\boxed{\nu = \sqrt{\frac{1-\beta}{1+\beta}}\nu'.} \tag{4}$$

If a source is moving away from you, $\beta > 0$ and $\nu < \nu'$. The light of the source is redshifted. If the source is moving towards you, replace β by $-\beta$, and the light is blueshifted. We show a spacetime diagram in Fig. 2 sketching the scenario for a source moving away. The figure is using SR units, so $c = 1$ (and hence the total time between flashes dt_R misses factors of c). To obtain the observed frequency between flashes, one needs to relate $dt_R = \gamma\Delta t'_R$.

2.2 Transverse effect

The expression we derived in the previous section holds for sources that are either moving towards you or away from you, the so-called longitudinal Doppler effect. What happens when a source is not moving exactly away from you, but at a slight angle Θ (see right side Fig. 2). In that case, our expression is simply modified by considering the projected velocity along the line of sight, $1 + v \rightarrow 1 + v\cos\Theta$. Inserting that into the expression for the longitudinal Doppler Eq. (3), we find

$$\nu = \gamma^{-1}(1+\beta\cos\Theta)^{-1}\nu'. \tag{5}$$

We can consider the extreme case when the source is moving transverse to the observer. In that case $\Theta = \pi/2$ and we find

$$\nu = \nu'\sqrt{(1-\beta^2)}. \tag{6}$$

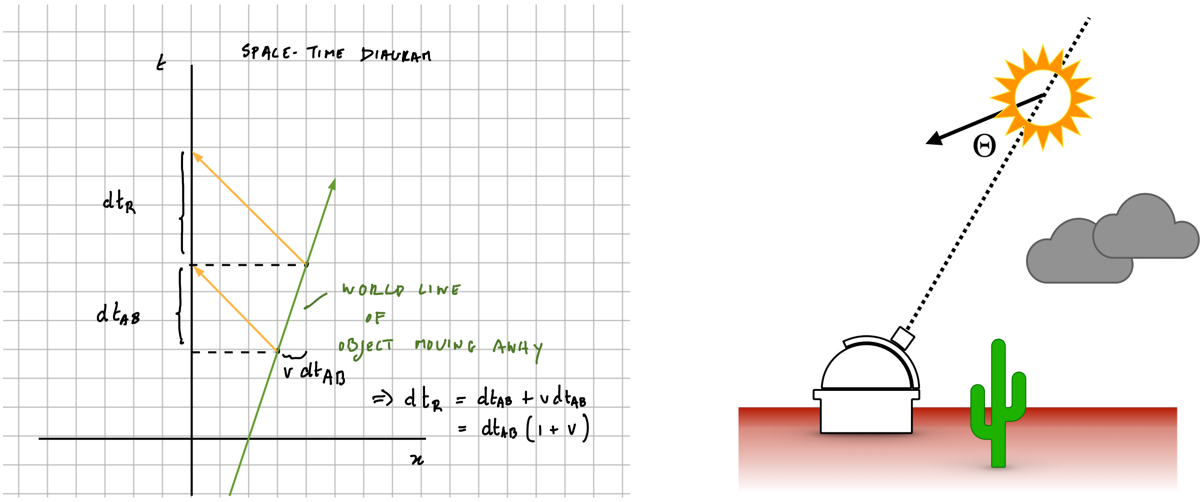

Figure 2: **Left:** Relativistic Doppler effect for a source emitting flashes at equal time intervals in frame S'. The flashes are seen by an observer in frame S at intervals that are either blueshifted (when the source is moving towards the observer) or redshifted (when the source is moving away, as shown in this diagram). **Right:** Doppler effects when the source is not exactly moving towards or away from the observer. With positive v denoting recession, we write $1 + v \to 1 + v\cos\Theta$.

Note that this is a second-order effect. It does not have an equivalent in classical theory. The above scenario is equivalent to 'Case 2' in Morin.

3 Rapidity

Recall that we could write the Lorentz transformations in matrix notation:

$$\begin{pmatrix} \Delta x' \\ \Delta t' \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta t \end{pmatrix}. \tag{7}$$

Now, let us define rapidity ϕ as

$$\tanh\phi \equiv \beta, \tag{8}$$

with $\beta = v/c$. There are some very nice properties of this function, which simplify SR transformations.

For example, consider Einstein's velocity transformations:

$$v_x = \frac{v'_x + \beta}{1 + \beta v'_x} = \frac{\tanh\phi_x + \tanh\phi}{1 + \tanh\phi\tanh\phi_x} = \tanh(\phi_x + \phi), \tag{9}$$

where the equality between the last two expressions is the consequence of the properties of the hyperbolic trig function (appendix A of Morin).

Similarly:

$$\gamma \equiv \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-\tanh^2\phi}} = \cosh\phi, \tag{10}$$

and

$$\gamma\beta = \sinh\phi. \tag{11}$$

This allows us to write the Lorentz transformation as

$$\begin{pmatrix} \Delta x' \\ \Delta t' \end{pmatrix} = \begin{pmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta t \end{pmatrix}, \tag{12}$$

which is remarkably similar to rotations around an angle α in Euclidean space:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \tag{13}$$

Instead of trig functions for rotations, we have hyperbolic trig functions.

How can this be useful? Suppose, for example, that you apply Lorentz transformations twice (see Q1 R3, 2023), once with v_1 (associated with rapidity ϕ_1) and another with v_2 (associated with rapidity ϕ_2). Doing this without rapidity is a bit of a mess, but as expected, it should lead to another Lorentz transformation, but now with a velocity that is transformed according to the Einstein velocity transformation. With rapidity, this can be seen easily since you would just apply the Lorentz transformation matrix twice and use the hyperbolic trigonometric identities to show that the new transformation leads to $v_{\text{final}} = \tanh(\phi_1 + \phi_2)$ which is identical to Eq. (9).

3.1 Interpretation

See Morin. When going from Eq. (11.62) to Eq. (11.63), you have to do two things. First, we should realize that locally, any function can be written as the function plus an infinitesimal displacement, which is determined by the first derivative multiplied by the displacement:

$$x(t+dt) = x(t) + \frac{\partial x}{\partial t}dt + \mathcal{O}(dt^2). \tag{14}$$

Here $\partial x/\partial t$ is the partial derivative, the derivative of a function keeping all other parameters fixed. But since v does not depend on anything but time, it can be replaced with the ordinary derivative below. To expand to first order in dt requires expanding the RHS of Eq. (11.62). We write:

$$\begin{aligned} \frac{v(t)+adt}{1+v(t)adt/c^2} &= (v(t)+adt)(1+v(t)adt/c^2)^{-1} \\ &= (v(t)+adt)(1-v(t)adt/c^2+\mathcal{O}(dt^2)). \end{aligned} \tag{15}$$

We thus find

$$v(t) + \frac{dv}{dt}dt = v(t) + adt(1-v(t)^2/c^2), \tag{16}$$

to first order in dt. This is equivalent to Eq. (11.63).

4 Relativity without c

Please read this section in Morin. It is a very nice derivation and quite powerful. It will not be something we will put in an exam (perhaps only a conceptual question).